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Compare permissions on Regular Files and Directories

Create a regular file named sample.txt and a directory named sample_dir. Create at least one file inside sample_dir.

Apply each of the following permissions using chmod. Perform the specified operations and record your observations.

Note: Perform the exercise as a normal user. Do not use sudo.

Permission	Regular File (sample.txt) – Test	Observation	Directory (sample_dir) – Test	Observation
400 (r--)	Try to read and modify the file	Read succeeds; modification fails.	Try to list and enter the directory	Listing succeeds; entering succeeds if directory has r but not x? Correction: listing succeeds, entering fails because x is required.
200 (-w-)	Try to read and modify the file	Read fails; writing/modification succeeds (subject to how the program opens the file).	Try to list, enter, and create a file	Listing fails; entering fails; creating files fails because x is required to access directory entries.
100 (--x)	Try to read and execute the file	Reading fails; execution may succeed only if the file is a valid executable/script and the required interpreter/access conditions are met.	Try to list and enter the directory	Listing fails; entering succeeds.
500 (r-x)	Try to read and modify the file	Reading succeeds; modification fails.	Try to list and enter the directory	Listing succeeds; entering succeeds.
600 (rw-)	Try to read and modify the file	Reading and modification succeed.	Try to list, enter, and create a file	Listing succeeds; entering fails; creating a file fails.
700 (rwx)	Try to read, modify, and execute the file	Reading and modification succeed; execution is allowed by the permission bits (assuming the contents are executable).	Try to list, enter, create, and delete files	All operations succeed.

1. 400 (r--) Try to read and modify the file __ Try to list and enter the directory

```
srinivas@DESKTOP-E4HFS1K:~$ whoami
srinivas
srinivas@DESKTOP-E4HFS1K:~$ pwd
/home/srinivas
srinivas@DESKTOP-E4HFS1K:~$ chmod 400 sample.txt
srinivas@DESKTOP-E4HFS1K:~$ chmod 400 sample_dir
srinivas@DESKTOP-E4HFS1K:~$ ls -l sample.txt
-r----- 1 srinivas srinivas 62 Aug 22 19:40 sample.txt
srinivas@DESKTOP-E4HFS1K:~$ ls -ld sample.txt
-r----- 1 srinivas srinivas 62 Aug 22 19:40 sample.txt
srinivas@DESKTOP-E4HFS1K:~$ cat sample.txt
hello srinivas
hello srinivas
hello again
final test
new text
srinivas@DESKTOP-E4HFS1K:~$ echo "new text" >> sample.txt
-bash: sample.txt: Permission denied
srinivas@DESKTOP-E4HFS1K:~$ ls sample_dir
ls: cannot access 'sample_dir/inside.txt': Permission denied
inside.txt
srinivas@DESKTOP-E4HFS1K:~$ cd sample_dir
-bash: cd: sample_dir: Permission denied
srinivas@DESKTOP-E4HFS1K:~$ cd ..
srinivas@DESKTOP-E4HFS1K:/home$
```

2. 200 (-w-) Try to read and modify the file ____ Try to list, enter, and create a file ____

```
srinivas@DESKTOP-E4HFS1K:~$ chmod 200 sample.txt
srinivas@DESKTOP-E4HFS1K:~$ chmod 200 sample_dir
srinivas@DESKTOP-E4HFS1K:~$ ls -l sample.txt
--w----- 1 srinivas srinivas 62 Aug 22 19:40 sample.txt
srinivas@DESKTOP-E4HFS1K:~$ ls -ld sample.txt
--w----- 1 srinivas srinivas 62 Aug 22 19:40 sample.txt
srinivas@DESKTOP-E4HFS1K:~$ cat sample.txt
cat: sample.txt: Permission denied
srinivas@DESKTOP-E4HFS1K:~$ echo "new text" >> sample.txt
srinivas@DESKTOP-E4HFS1K:~$ ls sample_dir/
ls: cannot open directory 'sample_dir/': Permission denied
srinivas@DESKTOP-E4HFS1K:~$ cd sample_dir/
-bash: cd: sample_dir/: Permission denied
srinivas@DESKTOP-E4HFS1K:~$ touch sample_dir/test.txt
touch: cannot touch 'sample_dir/test.txt': Permission denied
srinivas@DESKTOP-E4HFS1K:~$ cd ~
srinivas@DESKTOP-E4HFS1K:~$
```

3. 100 (--x) Try to read and execute the file ____ Try to list and enter the directory ____

```
srinivas@DESKTOP-E4HFS1K:~$ chmod 100 sample.txt
srinivas@DESKTOP-E4HFS1K:~$ chmod 100 sample_dir
srinivas@DESKTOP-E4HFS1K:~$ ls -l sample.txt
---x----- 1 srinivas srinivas 71 Aug 22 19:50 sample.txt
srinivas@DESKTOP-E4HFS1K:~$ ls -ld sample.txt
---x----- 1 srinivas srinivas 71 Aug 22 19:50 sample.txt
srinivas@DESKTOP-E4HFS1K:~$ cat sample.txt
cat: sample.txt: Permission denied
srinivas@DESKTOP-E4HFS1K:~$ ./sample.txt
bash: ./sample.txt: Permission denied
srinivas@DESKTOP-E4HFS1K:~$ ls sample_dir
ls: cannot open directory 'sample_dir': Permission denied
srinivas@DESKTOP-E4HFS1K:~$ cd sample_dir/
srinivas@DESKTOP-E4HFS1K:~/sample_dir$ cd ..
srinivas@DESKTOP-E4HFS1K:~$
```

4. 500 (r-x) Try to read and modify the file ____ Try to list and enter the directory ____

```
srinivas@DESKTOP-E4HFS1K:~$ chmod 500 sample.txt
srinivas@DESKTOP-E4HFS1K:~$ chmod 500 sample_dir
srinivas@DESKTOP-E4HFS1K:~$ ls -l sample.txt
-r-x----- 1 srinivas srinivas 71 Aug 22 19:50 sample.txt
srinivas@DESKTOP-E4HFS1K:~$ ls -ld sample.txt
-r-x----- 1 srinivas srinivas 71 Aug 22 19:50 sample.txt
srinivas@DESKTOP-E4HFS1K:~$ cat sample.txt
hello srinivas
hello srinivas
hello again
final test
new text
new text
srinivas@DESKTOP-E4HFS1K:~$ echo "new text" >> sample.txt
-bash: sample.txt: Permission denied
srinivas@DESKTOP-E4HFS1K:~$ ls sample_dir/
inside.txt
srinivas@DESKTOP-E4HFS1K:~$ cd sample_dir/
srinivas@DESKTOP-E4HFS1K:~/sample_dir$ cd..
cd..: command not found
srinivas@DESKTOP-E4HFS1K:~/sample_dir$ _
```

5. 600 (rw-) Try to read and modify the file ____Try to list, enter, and create a file ____

```
srinivas@DESKTOP-E4HFS1K:~$ chmod 600 sample.txt
srinivas@DESKTOP-E4HFS1K:~$ chmod 600 sample_dir/
srinivas@DESKTOP-E4HFS1K:~$ ls -l sample.txt
-rw----- 1 srinivas srinivas 71 Aug 22 19:50 sample.txt
srinivas@DESKTOP-E4HFS1K:~$ ls -ld sample.txt
-rw----- 1 srinivas srinivas 71 Aug 22 19:50 sample.txt
srinivas@DESKTOP-E4HFS1K:~$ cat sample.txt
hello srinivas
hello srinivas
hello again
final test
new text
new text
srinivas@DESKTOP-E4HFS1K:~$ echo "hello again" >> sample.txt
srinivas@DESKTOP-E4HFS1K:~$ ls sample_dir
ls: cannot access 'sample_dir/inside.txt': Permission denied
inside.txt
srinivas@DESKTOP-E4HFS1K:~$ cd sample_dir
-bash: cd: sample_dir: Permission denied
srinivas@DESKTOP-E4HFS1K:~$ cd ..
srinivas@DESKTOP-E4HFS1K:/home$ touch sample_dir/test.txt
touch: cannot touch 'sample_dir/test.txt': No such file or directory
srinivas@DESKTOP-E4HFS1K:/home$ ls
srinivas
srinivas@DESKTOP-E4HFS1K:/home$ cd srinivas/
srinivas@DESKTOP-E4HFS1K:~$ touch sample_dir/test.txt
touch: cannot touch 'sample_dir/test.txt': Permission denied
srinivas@DESKTOP-E4HFS1K:~$
```

6. 700 (rwx) Try to read, modify, and execute the file ____Try to list, enter, create, and delete file _____

```
srinivas@DESKTOP-E4HFS1K:~$ chmod 700 sample.txt
srinivas@DESKTOP-E4HFS1K:~$ chmod 700 sample_dir/
srinivas@DESKTOP-E4HFS1K:~$ ls -l sample.txt
-rwx----- 1 srinivas srinivas 83 Aug 22 19:57 sample.txt
srinivas@DESKTOP-E4HFS1K:~$ ls -ld sample.txt
-rwx----- 1 srinivas srinivas 83 Aug 22 19:57 sample.txt
srinivas@DESKTOP-E4HFS1K:~$ cat sample.txt
hello srinivas
hello srinivas
hello again
final test
new text
new text
hello again
srinivas@DESKTOP-E4HFS1K:~$ echo "hello again" >> sample.txt
srinivas@DESKTOP-E4HFS1K:~$ ./sample.txt
./sample.txt: line 1: hello: command not found
./sample.txt: line 2: hello: command not found
./sample.txt: line 3: hello: command not found
./sample.txt: line 4: final: command not found
./sample.txt: line 5: new: command not found
./sample.txt: line 6: new: command not found
./sample.txt: line 7: hello: command not found
./sample.txt: line 8: hello: command not found
srinivas@DESKTOP-E4HFS1K:~$ ls sample_dir/
inside.txt
srinivas@DESKTOP-E4HFS1K:~$ cd sample_dir/
srinivas@DESKTOP-E4HFS1K:~/sample_dir$ touch newfile.txt
srinivas@DESKTOP-E4HFS1K:~/sample_dir$ ls
inside.txt  newfile.txt
srinivas@DESKTOP-E4HFS1K:~/sample_dir$ rm newfile.txt
srinivas@DESKTOP-E4HFS1K:~/sample_dir$ ls
inside.txt
srinivas@DESKTOP-E4HFS1K:~/sample_dir$ cd ..
srinivas@DESKTOP-E4HFS1K:~$
```

Questions:

1. What does read (r) permission allow you to do with a regular file?

It allows you to **view/read the contents** of the file.

2. What does read (r) permission allow you to do with a directory?

It allows you to **read the directory's contents/list of filenames**. It does not by itself allow you to enter or access those files.

3. What does write (w) permission mean for a regular file?

It allows you to **modify, overwrite, or write to the file**. It does not automatically allow you to read its contents.

4. What does write (w) permission mean for a directory?

It allows you to **add, remove, or rename directory entries**, but normally you also need x permission to actually perform these operations.

5. What does execute (x) permission mean for a regular file?

It allows the file to be **executed as a program**, provided its contents and execution environment support that.

6. Why is execute (x) permission important for accessing a directory?

Directory x means **search/traverse** permission. It allows you to enter the directory and access entries within it when the relevant permissions are also available.

7. Which permission combination allows the owner to fully access a regular file?

700 (rwx) gives the owner read, write, and execute permission.

8. Which permission combination allows the owner to fully access a directory?

700 (rwx) gives the owner permission to list, enter/traverse, create, delete, and modify directory entries.

9. Why do the same r, w, and x permissions behave differently for a regular file and a directory?

Because a regular file and a directory represent different filesystem objects. For a **file**, r controls its contents, w controls modification of its contents, and x controls execution. For a **directory**, r controls reading its list of entries, w controls changing its entries, and x controls traversal/access through the directory. Thus, the same permission bits have meanings appropriate to the type of filesystem object.